



Occupational health

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Out line

Introduction

Definition of
occupational
health nursing

Theatrical frame
work.

Standards and
objective of
Occupational
Health Nursing
Practice

Out line continue



objectives

At the end of
the presentation
the student will
be able to :

Define
occupational
health nursing

Discuss the
Theatrical frame
work of
occupational
health nursing
practice

Discuss the
standards of
Occupational
Health Nursing
Practice

List the
environmental
work factors and
hazards which
affect health:

Mention the
controlling
measures of
occupational
health hazards

enumerate
occupational
health program.

Mention the
role of
Occupational
Health Nurses



“ What the mind
does not know,
the eyes do not
see”

introduction

Work is basic to life throughout the world, most adults spend almost one third of their time at work

The age of worker in Egypt ranges from 15 to 64 years, An estimated 22570 occupational injures, resulted in 127 fatalities, 366 cases of handicapped and 56 disability that lead to 459779 days of absenteeism **(Central Agency for Mobilization and Public Statistics, 2003).**





Definition of occupational health nursing



- is defined as the specialty practice that focuses on the promotion, prevention, and restoration of health within the context of a safe and healthy environment. It includes the prevention of adverse health effects from occupational and environmental health and safety programs and services to clients.



Theatrical frame work:

Occupational health nursing derives its theoretical, conceptual and factual framework from:

- **Nursing science**
- **Medical science**
- **Occupational health sciences**
- **Epidemiology**
- **Business and economic**
- **Environmental health**
- **Legal and ethical issues;**



Objective of occupational health

- Reduce death from work–related injuries.
- Reduce work-related injuries resulting in medical treatment, lost time from work, or restricted work activity.
- Reduce the rate of injury and illness cases involving days away from work due to overexertion or repetitive motion.
- Reduce pneumoconiosis deaths.

- Reduce work–related assault and reduce deaths from work–related homicides.
- Reduce the number of persons who have elevated bloods' lead level concentrations from work exposures.
- Reduce occupational skin diseases or disorders among full-time workers.
- Increase the proportion of worksites employing 50 or more persons that provide programs to prevent or reduce employee stress.
- Reduce occupational needle stick injuries among healthcare workers.
- Reduce new cases of work – related noise induced hearing loss.



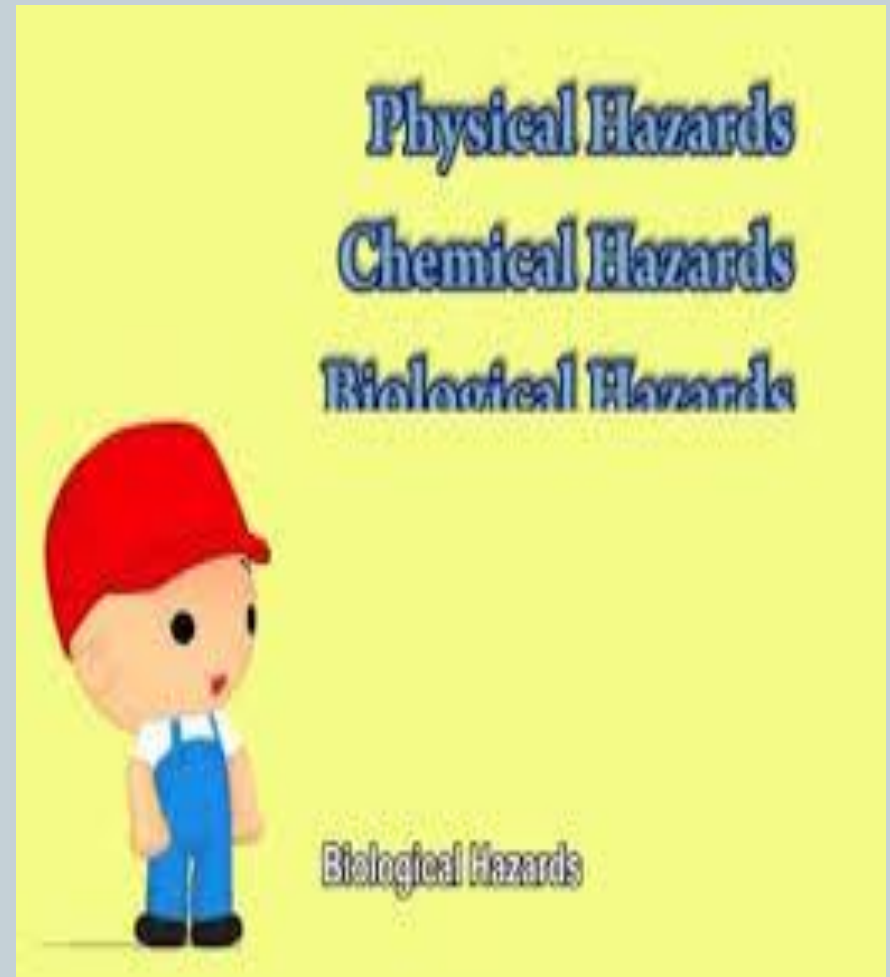
Standards of occupational health nursing



- Standard 1: Assessment
- Standard 2:
 - Diagnosis
- Standard 3:
 - Outcome
- Standard 4:
 - Planning
- Standard 5:
 - Implementation
- Standard 6:
 - Evaluation
- Standard 7:
 - Resources management
- Standard 8:
 - Professional development
- Standard 9:
 - Collaboration
- Standard 10:
 - Research
- Standard 11:
 - Ethics

Types of occupational hazards and its health effects

- Physical hazards
- Chemical hazards
- Biological hazards
- Psychological hazards
- Ergonomic hazards
- Accidents



Physical hazards

- Physical hazards are agents or forces inherent from the nature of a work environment, or process that may cause tissue damage or other physical harm. The various features that make up the physical work environment includes work space, temperature, lightening, noise, vibration, radiation, soundness of the building, the equipment, magnetic fields, and microwaves.





Physical hazards

- **A)-Noise:**

Noise is the most prevalent work place hazard in the world it's effect extended beyond the workers operating the machine to all those in the area. One worker in four exposed to 90 DBA noise over a working life time develop a hearing loss that can be attributed to occupational noise exposure

- ***B) Heat:***

heat is a potential physical hazard that can be existed in almost any work place, especially during summer. It is a particular hazard especially in tropical climates. Thermal stress experienced when working in condition of excessive heat or cold that can lead to multiple pathologies, including cardiovascular, metabolic disturbances, central/peripheral neurological alterations, and mental status changes, resulting in impaired judgment/performance and increased risk or accidents

Chemical hazards

- **(2)-Chemical factors:**

It is the chemical agents present in the work environment that may threaten worker health and safety, an estimated 32 million workers are annually exposed to one or more chemical hazards that may cause serious and adverse health effects(as brain, blood,kidney,liver,bone,and central nervous system)

- **In work place people absorb chemicals by three main routes:**

The respiratory system - the skin and/or the gastrointestinal tract,



Sources of chemical substances

the American national standards institute classifies chemicals

- Dusts particulates, fumes, mists, vapors, and gases.
- Chemical formation includes solutions, metals, and solvents.
- Aerosols, pharmaceuticals, oils, synthetic textiles, pesticides, and explosives.
- Commercial products are often formulated with various additives and stabilizers to render a useable form of substances that may be toxic.



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Hazard warnings



Toxic



Harmful/
Irritant



Corrosive



Explosive



Oxidizing



Highly flammable



Ergonomic hazards

- Ergonomic factors include all the interactions between the worker, the demands of the job, the work setting, and the overall environment.
- Ergonomics (some times called human engineering) they include all the physiologic and psychological demands on the job as well as other work place stressors that can cause anxiety. These can include physical conditions in a workspace such as the design of necessary tools equipment, tool design, and frequent repetition of a limited movement, lighting or ventilation, lack of toilet facilities and/or drinking water.



(4) -Biological factors:

- Biological factors are living organisms found in the work environment; these include bacteria, virus's, and fungi, parasites of various types, Insects, animals and even toxic plants.
- Potential hazards, such as infectious or parasitic diseases, may derive from exposure to biologic agents can be direct or indirect, and occurs through one of three routes of transmission air borne, droplet or contact
- As hepatitis, HIV, rubella,tetanus,salmonella.

5) - Psychosocial factors:

- Psychological stress can result from personal problems such as a terminally ill spouse, a painful divorce or other family difficulties or crises such problems can result in depression or even despair. These types of personal dilemmas influence the quality and quantity of work produced and worker safety.





Accidents

- An accident is an unanticipated, sudden event that may become an undesired outcome such as property damage, badly injury, or death. Injury is physical damage to body tissues caused by an accident or by exposure to environmental stressors, much work injuries result from accidents, such as a heavy object falling and crushing bones in a worker's foot.

Controlling occupational health hazards

- three lines of defense against health hazards. These are identified as:
- Engineering controls.
- Personal protective equipment.
- Administrative or work practice controls.



1 - Engineering controls:



Engineering controls deal directly with the hazards by removing it and suppressing it, it is physical or mechanical devices that isolate or remove health hazards from the work place, engineering controls must be properly maintained or replaced as required to ensure their effectiveness.

- Personal protective clothing and equipment

Personal protective clothing and equipment must be used whenever occupational exposure remains after instituting engineering and work practice control as :

- **Gloves :**

There are three main types of gloves, clean latex gloves, sterile gloves, heavy duty and utility gloves. It is used for barrier protection. Surgical aspects are made of latex.

- **Helmet :**

Helmet or other forms of head protection may have to be compatible with other devices such as gas masks. The helmet also forms a useful base to which other equipment can be attached



- **Firm grasp handles and foot protection devises**

- **Ear protection :**

Is widely used to reduce the incidence of hearing loss, three types of ear protection are available, ear muffs or defenders, glass down for making into ear inserts and ear plugs .



- **Eyes protections devices :**

Flying objects are another cause of struck by injuries. Small airborne particles released during operations such as grinding, chipping, and machining can cause severe eye injuries and blindness



occupational health program



- Health promotion through fulfilling requirement of physical, mental and social well being
- Prevention of health hazards
- Control of health hazards.
- rehabilitation



component of occupational health program:

health promotion through

- socioeconomic promotion
- adequate nutrition
- sanitary environment
- medical health promotion

prevention of occupational hazards

- prevent occupational injury
- prevent occupational disease



health education

rehabilitation

health appraisal

- comprehensive medical examination
- screening test
- curative services
- health record

Members of occupational health team include

Two professionals that provide on-site occupational health services are, the occupational health nurse and the safety engineer other members of the inter disciplinary team may be include an industrial hygienist, epidemiologist, toxicologist, and occupational physician, therefore the position of the occupational health nurse is the corner stone of the occupational health. Finally, the occupational health team is not completed without the workers themselves.

Role of Occupational Health Nurses

- The OHN's responsibilities have expanded immensely to encompass not only the responsibilities previously listed but also a wide range of job duties, including but not limited to:
- **Case management**
- **Educator**
- **Worker and workplace hazard detection**
- **Counseling and crisis intervention**
- **Health promotion**
- **Administrator**
- **Researcher**
- **Clinicians and practitioner**

- **Clinician / Practitioner:**



Assesses the health problem by gathering data about the worker and the work environment. Specific functions of the occupational health nurse clinician/practitioner include but are not limited to the following activities:

- 1- Assesses the work environment for actual or potential health and safety hazards.
- 2- A assesses the health status of workers.
 - a) - Health history.
 - b) - Occupational health history.
 - c) - Lifestyle -health risk appraisal.

- 3 - Performs physical examinations.
- a) -Job placement.
- b) -Periodic.
- c) -Return to work.
- d)- Health surveillance.
- e) - Retirement.
- 4 - Conducts appropriate laboratory tests.
- 5 - Provides direct nursing care for occupational and no occupational injuries and illnesses.

- 6 - Conducts health education and counseling.
- 7 - Collaborates, communicates, and consult with other members of the occupational health team.
- 8- Institutes appropriate personal protection programs.
 - a) Safety glasses.
 - b) Safety shoes.
 - c) Hearing protection.
 - d) Respiratory protection.
 - e) Personal protective clothing.

- 9 - Maintains accurate and complete health records of the workers.



- 10 - Develops and implements programs designed to correct or minimize the identified health and safety hazards.
- 11- Initiates referrals to community specialists.
- 12-Conducts screening programs.
- 13-Develops and implements health promotion and\or disease prevention programs.
- 14-Evaluates the various programs





حما الله عمالنا --- حفظ الله أياديهم ----
وحفظ الله وطننا
واليد التي تبني هي التي تستحق التقدير.

